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# ECSE 4660/6660: Internetworking of Things

Lecture 11: Link Layer  
Fundamentals – Part III: BLE

Ish Jain  
ECSE, RPI

Slides Acknowledgement [Pannuto, Ghena, Campbell] from UCSD

Spring 2026

# Administrative

- Use Piazza for any questions
  - [https://piazza.com/rpi/spring2026/ecs\\_e46606660](https://piazza.com/rpi/spring2026/ecs_e46606660)
- Today's Quiz is available on **LMS**
  - Multiple choice questions
  - Due tonight 11:59 PM
- Lab
  - **TA hours 4-5 PM** daily in 5312 lab
  - Lab 2 late due **Fri, 2/20**
  - Lab 3 due **Fri, 2/27**
- HW2 on **LMS** **due Mon 2/23**
  - No Late submission allowed
- **Project group formation**
  - [IoT\\_SP26\\_project.xlsx](#)
- **Project Next Steps**
  - Work with TA to get the hardware needed
  - Mid-term report **due 3/20**
  - In-class presentation+2min video 4/20, 4/23
  - Live-demo 4/27
  - Final code/report 5/1

# Dissecting 5G in the Wild: Performance, Coverage, and Support for Next-Gen Applications

## Moinak Ghoshal

Ph.D. Student, Institute for the Wireless Internet of Things  
Northeastern University

ECSE Topical Seminar  
JEC 3117  
Thu, February 19, 2026 at 4:00 PM

Today

Since its rollout, 5G has promised unprecedented data rates and ultra-low latency, with the potential to finally enable the next generation of bandwidth-intensive and latency-critical applications such as AR/VR, panoramic video streaming, cloud gaming, and connected autonomous vehicles, often dubbed as 5G “killer apps”. Yet, realizing this promise in practice has proven challenging. The use of higher frequency bands, while ensuring higher throughput, comes at the cost of reduced range and greater susceptibility to performance fluctuations. Moreover, the introduction of multiple frequency bands, the coexistence with LTE, and the availability of different deployment modes (Non-Standalone and Standalone 5G) have made the cellular ecosystem increasingly complex. While testbeds offer valuable insights, they often fail to capture the full range of real-world scenarios, as different network operators implement unique policies and configurations that are difficult to replicate in controlled environments.

Moinak Ghoshal is a final-year Ph.D. student at Northeastern University’s Institute for the Wireless Internet of Things, advised by Prof. Dimitrios Koutsonikolas. His research interests lie in experimental wireless networking and mobile computing, with a current focus on evaluating 5G/NextG networks and enabling an emerging class of applications over them. Outside academia, he has collaborated with leading industry research labs such as HPE Labs, AT&T Labs, InterDigital, Futurewei, Rakuten, and Ookla. He received the Best Dataset Award at PAM 2021 and was a runner-up for the Community Contribution Award at ACM CoNEXT 2025. His research has been published in leading venues such as ACM IMC, ACM SIGMETRICS, ACM CoNEXT, IEEE INFOCOM, and IEEE/ACM Transactions on Networking.



# References

## 1. Bluetooth Low Energy (BLE)

- Reference

N. Hunn, “**Essentials of Short-Range Wireless**,” Cambridge University Press, 2010.

Chapter 7: BLE

## 2. IEEE 802.15.4 and Zigbee

- Reference

N. Hunn, “**Essentials of Short-Range Wireless**,” Cambridge University Press, 2010.

Chapter 6: IEEE 802.15.4 and ZigBee

cellular 3G

CDMA (following up from Lec 10)



OFDM

# CDMA



## ■ Recall:

- FDMA: each user occupies a frequency-slot throughout the communication session
- TDMA: each user occupies a time-slot throughout the communication session

## ■ CDMA



- Multiple users use the same time and frequency slots at the same time!
  - A user is differentiated by a code that acts as the key to identify that user
  - Orthogonal codes are assigned such that when they are used at the same time in the same band, a receiver knowing the code of the particular user can detect that user signal among all simultaneously received signals from other users
  - Here also, two versions exist:
    - CDMA/FDD: forward and reverse channels use different carriers
    - CDMA/TDD: forward and reverse channels use same carrier
- 

# CDMA Illustration

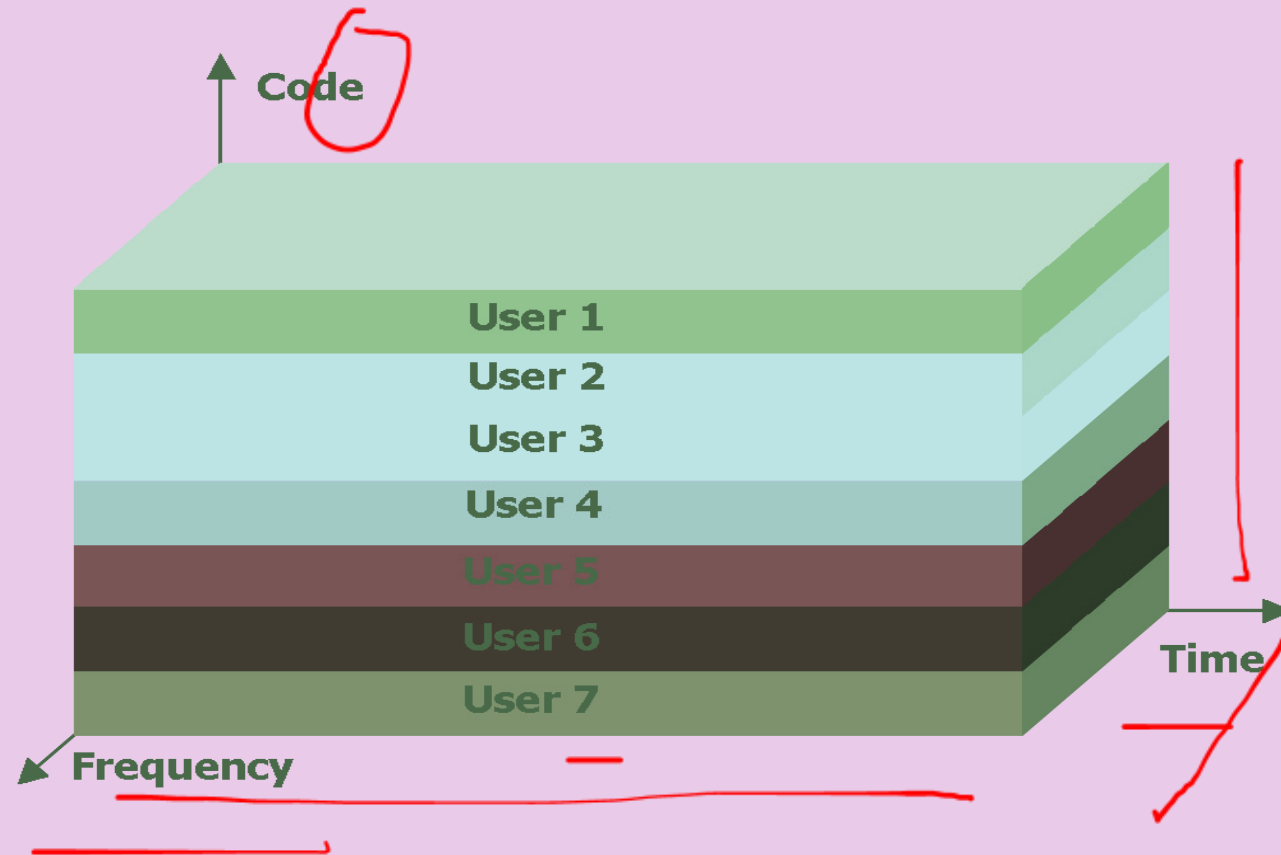


Figure 6: Simple illustration of CDMA

# CDMA/FDD and CDMA/TDD Illustration

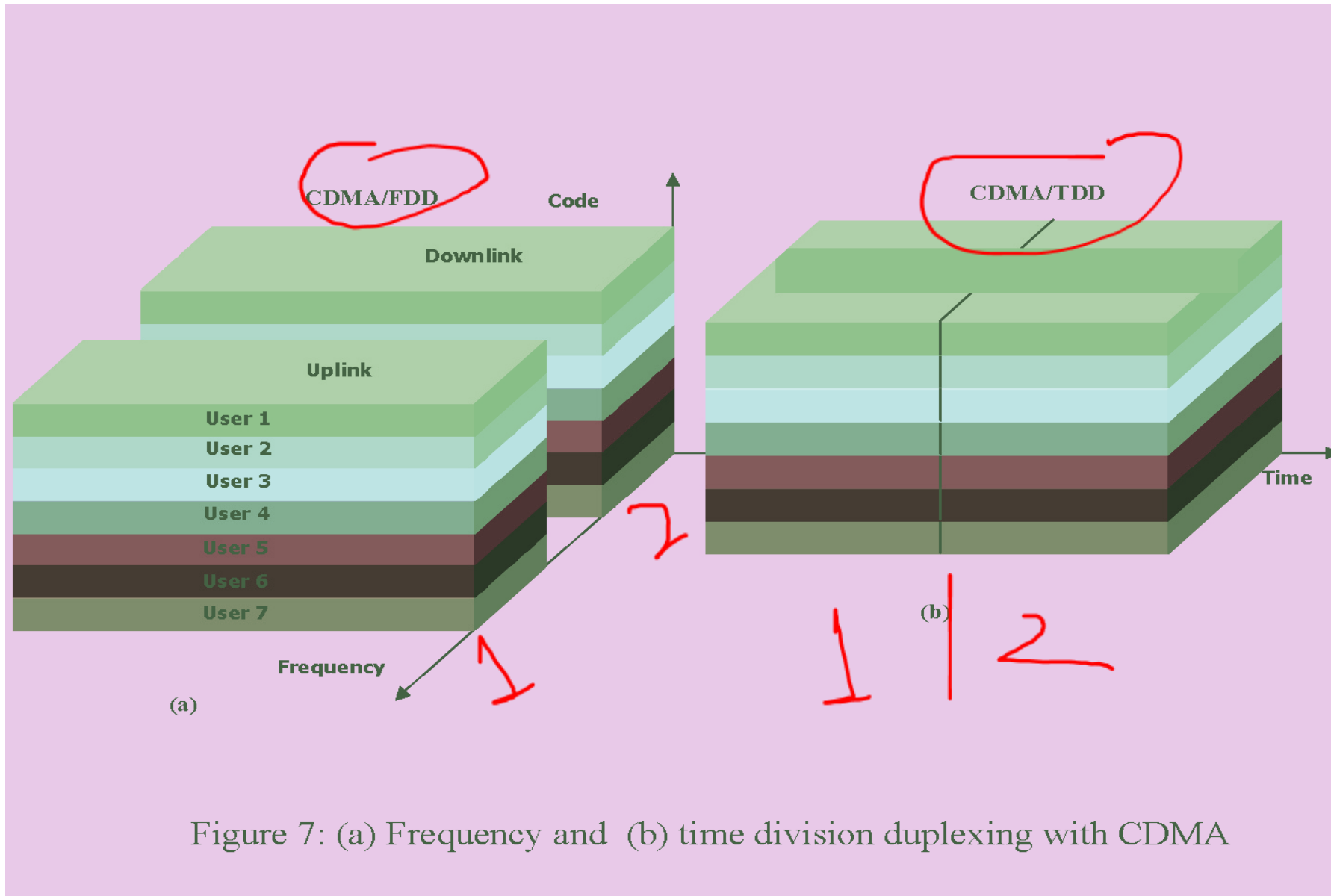


Figure 7: (a) Frequency and (b) time division duplexing with CDMA



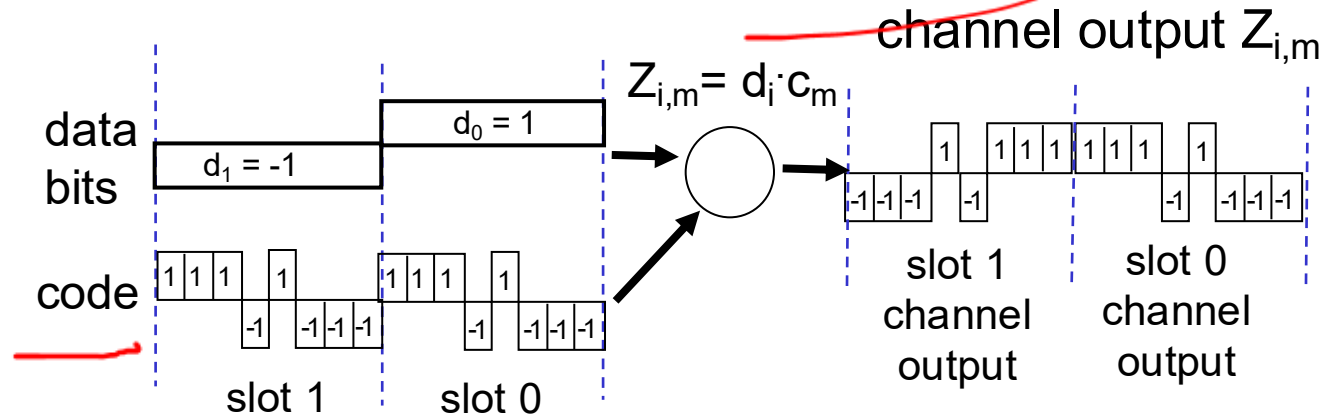
# Code Division Multiple Access (CDMA)

- unique “code” assigned to each user; i.e., code set partitioning
  - all users share same frequency, but each user has own “chipping” sequence (i.e., code) to encode data
  - allows multiple users to “coexist” and transmit simultaneously with minimal interference (if codes are “orthogonal”)
- **encoding**: inner product: (original data) X (chipping sequence)
- **decoding**: summed inner-product: (encoded data) X (chipping sequence)

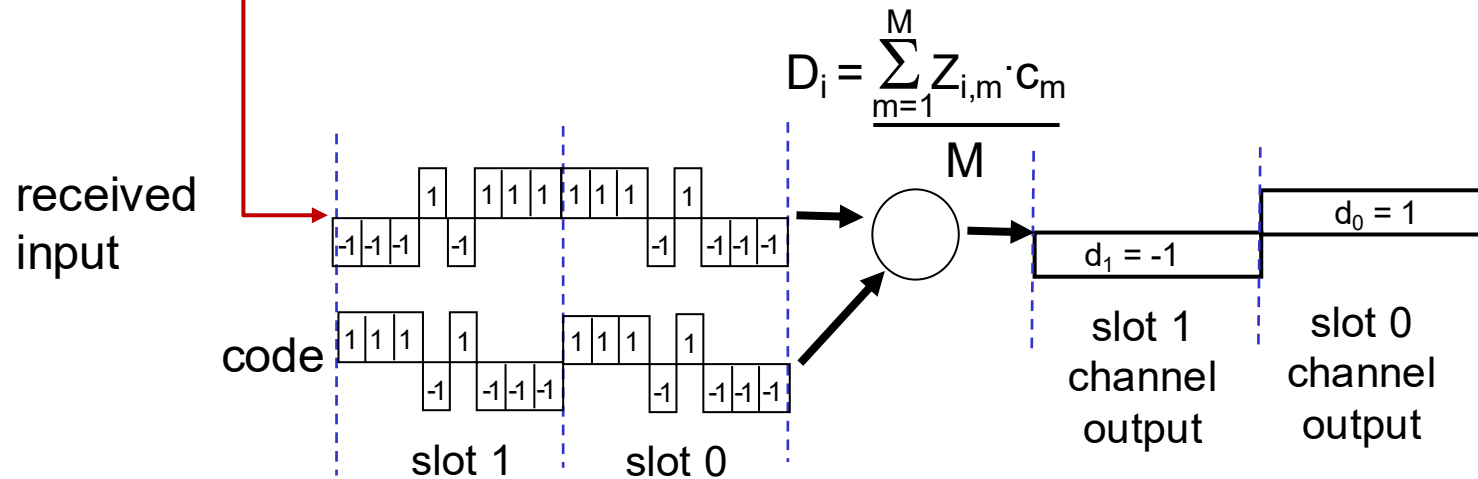
# CDMA encode/decode

$$M = 8$$

sender



receiver

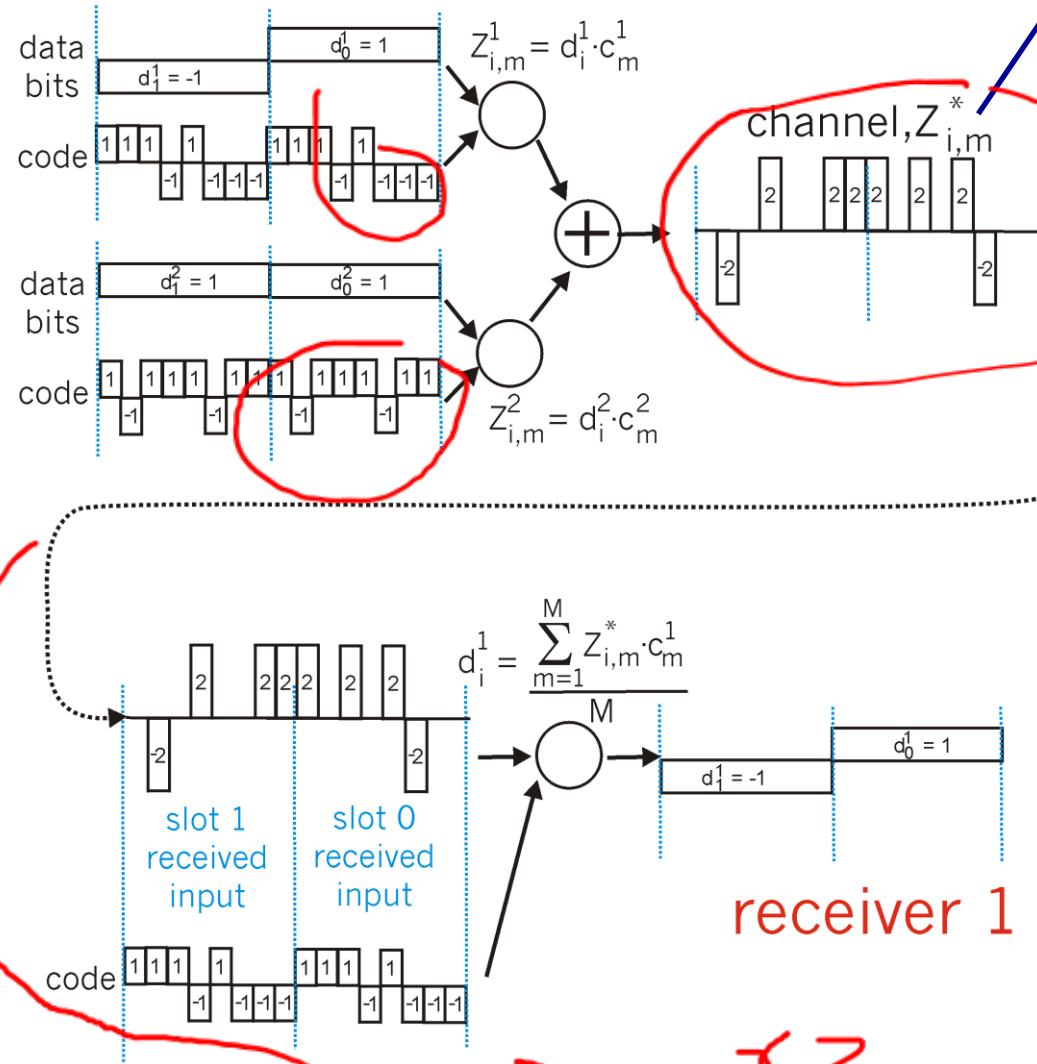


... but this isn't really useful yet!

# CDMA: two-sender interference

Sender 1

Sender 2



channel sums together transmissions by sender 1 and 2

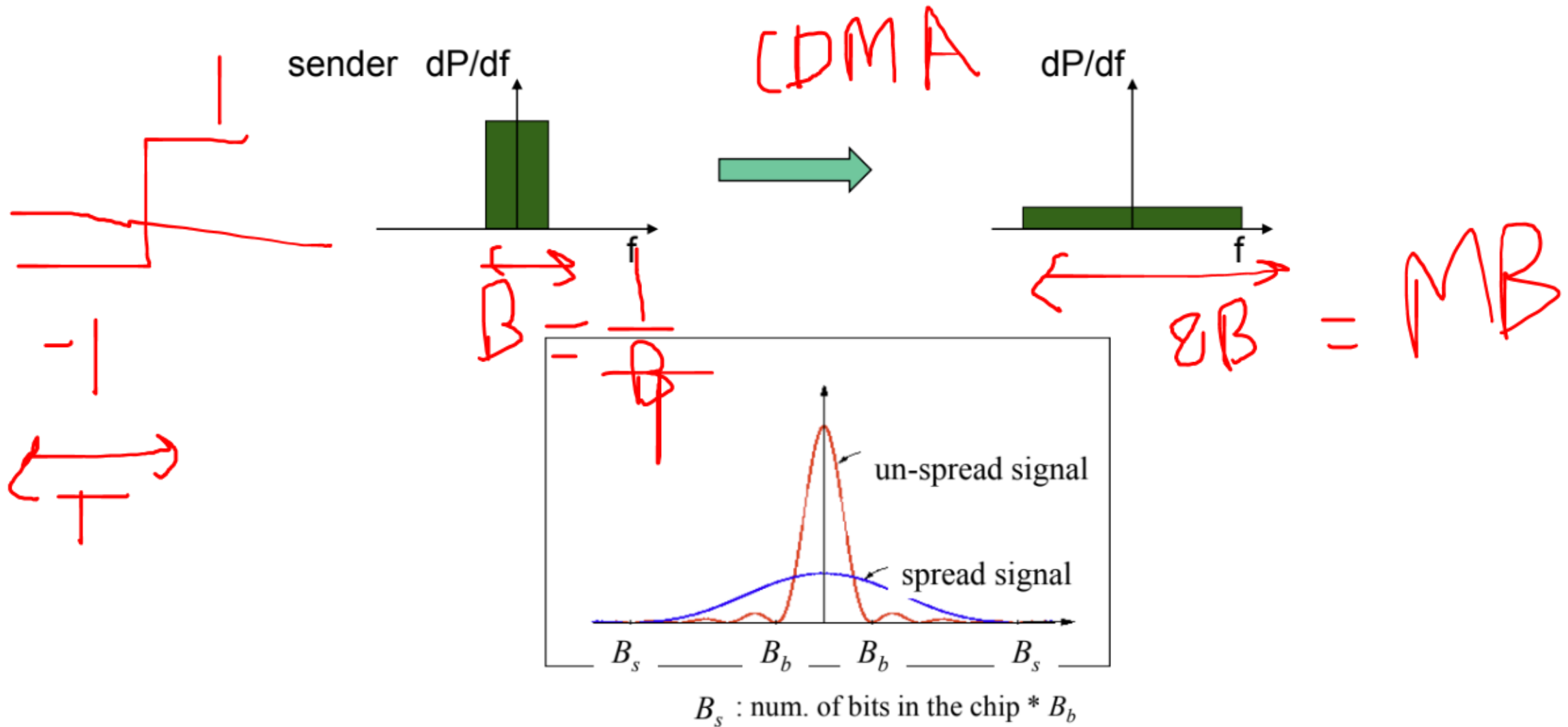
$$\sum_m c_m^1 c_m^2 = 0$$

using same code as sender 1, receiver recovers sender 1's original data from summed channel data!

... now *that's* useful!

82

# Effect of spreading: frequency-domain perspective



# Problem (similar to the HW2 problem)

A CDMA system operates with:

- **User Data Rate:** 200 kbps
- **Chipping Sequence Length:** 10
- **Total Users:** 6

**Question:** What is the **spectral efficiency** of the system in bits/sec/Hz?

B  
M

0--

6B  
-----  
MB

# Bluetooth Low Energy (BLE)

# Bluetooth Low Energy Resources

- Good walkthrough of BLE:
  - <https://www.silabs.com/documents/public/user-guides/ug103-14-fundamentals-ble.pdf>
- [[5.2 specification](#)] [[4.2 specification](#)] (link to PDF download)
  - Also: [[Supplement v9](#)]
- More boots-on-the-ground view:
  - <https://inst.eecs.berkeley.edu/~ee290c/sp18/lec/Lecture7A.pdf>
    - From a team that has implemented BLE HW several times
  - [https://download.ni.com/evaluation/rf/intro\\_to\\_bluetooth\\_test.pdf](https://download.ni.com/evaluation/rf/intro_to_bluetooth_test.pdf)

# Outline

- **BLE Background**

- **BLE Layers**

- Physical Layer
  - Link Layer
- 

- **BLE roles**

- Advertising
  - Scanning
- 



# Bluetooth Classic vs BLE

## Bluetooth Classic (around since 1999)

- Continuous data communication
- Can afford higher battery power requirements



Headphones



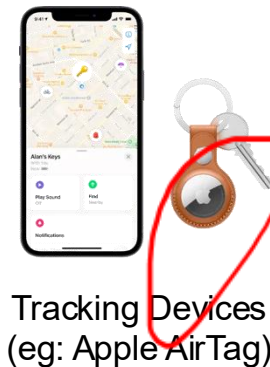
Car BT Audio



Mobile Photo Printers



Fitness Trackers



Tracking Devices  
(eg: Apple AirTag)



Switches  
and Sensors

## Bluetooth Low Energy (introduced in v4.0)

- Small amounts of data transfer
- Low power requirements
  - “months or years” on small batteries
- Ideal for IoT devices

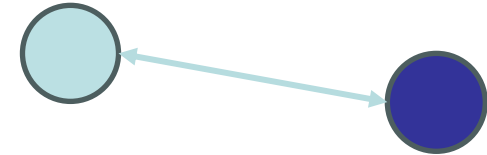
# Bluetooth has a long history — the IoT is near-exclusively BLE (Bluetooth 4.0+)

| Year | Bluetooth Standard | Data Rate                  | Modulation                       | Notes                                                                                                                                       |
|------|--------------------|----------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1999 | V1.0               | 1 Mb/s                     | GFSK                             | • The Bluetooth 1.0 Specification is released by the Bluetooth SIG                                                                          |
| 2003 | V1.2               | 1 Mb/s                     | GFSK                             | • First FDA-approved Bluetooth medical system. Bluetooth product shipments grow to 1 million/week                                           |
| 2004 | V2.0 + EDR         | 1 Mb/s<br>2 Mb/s<br>3 Mb/s | GFSK<br>$\pi/4$ -DQPSK<br>8-DPSK | • Introduction of Enhanced Data Rate (EDR) for faster data transfer.<br>• Bluetooth product shipments surpasses to 3 million/week           |
| 2007 | V2.1 + EDR         | 1 Mb/s<br>2 Mb/s<br>3 Mb/s | GFSK<br>$\pi/4$ -DQPSK<br>8-DPSK | • Introduction of secure simple pairing (SSP) and extended inquiry response (EIR) for Bluetooth devices                                     |
| 2009 | V3.0+HS            | 1 Mb/s<br>2 Mb/s<br>3 Mb/s | GFSK<br>$\pi/4$ -DQPSK<br>8-DPSK | • Introduction of AMP (Alternative MAC/PHY) and the addition of 802.11 as a high-speed transport with data transfer speeds up to 24 Mbit/s. |

|      |              |                            |                                  |                                                                                                                                                                                                                                                    |
|------|--------------|----------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2009 | V3.0+HS      | 1 Mb/s<br>2 Mb/s<br>3 Mb/s | GFSK<br>$\pi/4$ -DQPSK<br>8-DPSK | • Introduction of AMP (Alternative MAC/PHY) and the addition of 802.11 as a high-speed transport with data transfer speeds up to 24 Mbit/s.                                                                                                        |
| 2010 | V4.0 (Smart) | 1 Mb/s<br>2 Mb/s<br>3 Mb/s | GFSK<br>$\pi/4$ -DQPSK<br>8-DPSK | • Introduction of Bluetooth Low Energy protocol and AES encryption                                                                                                                                                                                 |
| 2013 | V4.1         | 1 Mb/s<br>2 Mb/s<br>3 Mb/s | GFSK<br>$\pi/4$ -DQPSK<br>8-DPSK | • MWS (Mobile Wireless Standard) Coexistence<br>• SIG membership surpasses 20,000 companies                                                                                                                                                        |
| 2014 | V4.2         | 1Mb/s<br>2Mb/s<br>3Mb/s    | GFSK<br>$\pi/4$ -DQPSK<br>8-DPSK | • Smart sensor allows flexible internet connectivity<br>• Increased privacy (Le Privacy 1.2 and LE Secure Connections)<br>• LE Data Length Extension increases data throughput with packet capacity increase of 10x compared to previous versions. |

# Basics of Bluetooth Low Energy (BLE)

- Direct device-to-device communication
  - Usually: Computer to Thing
  - Smartphone to device, Laptop to device, etc.
- Focus on making the “Thing” really low energy
  - Push energy-intensive requirements onto “Computer”
- Devices (Computer or Thing) are servers with accessible fields
  - Not the traditional send-explicit-packets interface you might be expecting
  - Lower layers are still exchanging packets to make it work



# A note on outdated notation

- Master/Slave paradigm
  - Master is the “Computer” and is in charge of interaction
  - Slave is the “Device” and has little control over interaction parameters
- Field is changing for the better. It’s going to take some time.
  - Central/Peripheral
  - Device/Peripheral
  - Controller/Peripheral
  - Primary/Secondary

# Bluetooth Specification

- Problem: a bit overwhelming...
  - 5.2 spec: 3256 pages
  - We only care about Vol 6: Low Energy Controller
    - Part A: Physical Layer Specification
    - Part B: Link Layer Specification
    - CSS: Part A: Data Types Specification
    - So ~250 pages
- Tip: be willing to just ignore things when skimming specs
  - 5.2 spec covers BLE and Bluetooth Classic and a bunch of upper layer stuff that we never have to care about

# BLE mechanisms

## ■ Advertising

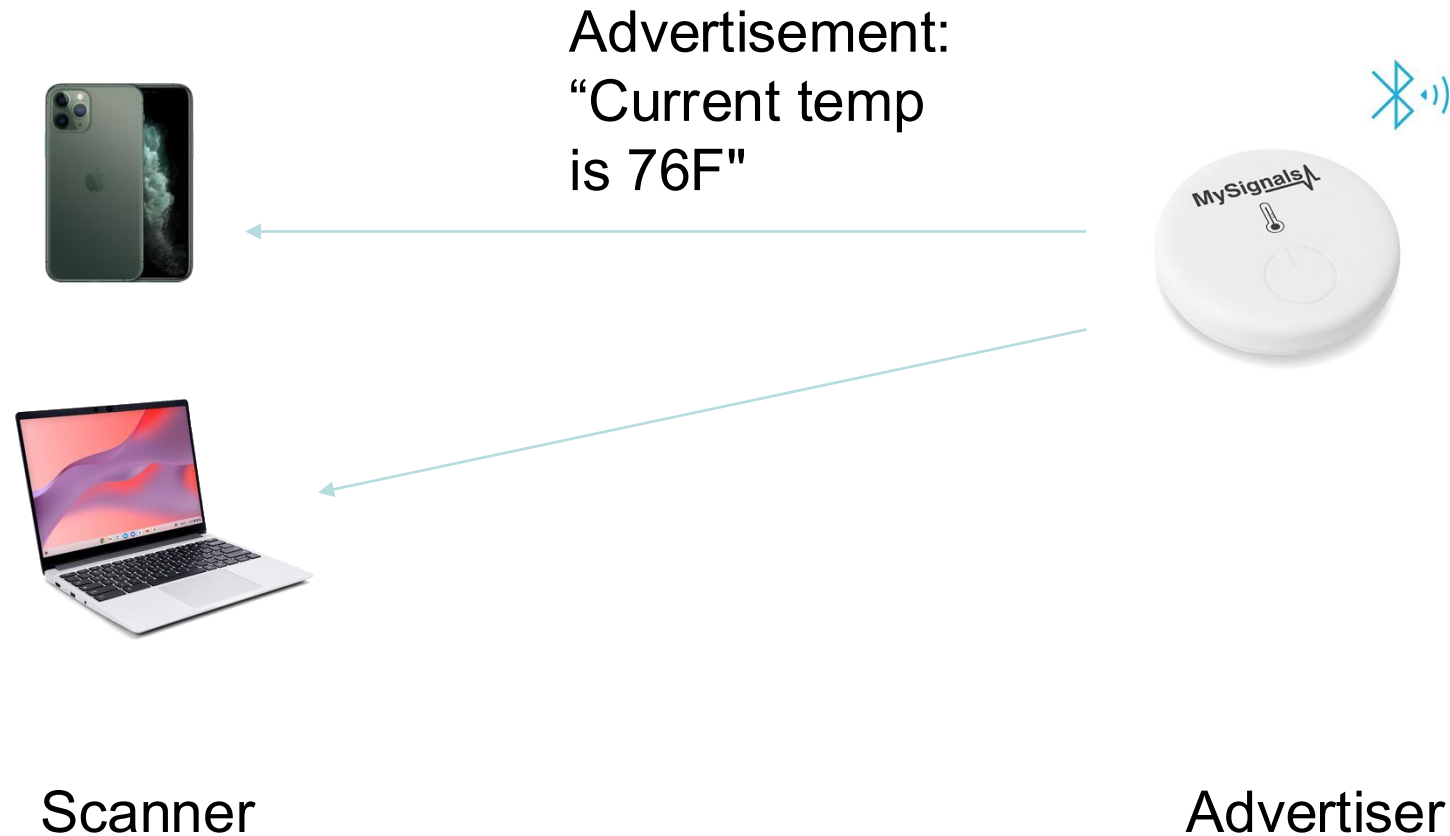
- Discovery
- Advertisements – broadcast messages indicating device details
- Ephemeral, uni-directional communication from Advertiser to Scanner(s)
- ALOHA access control

## ■ Connections

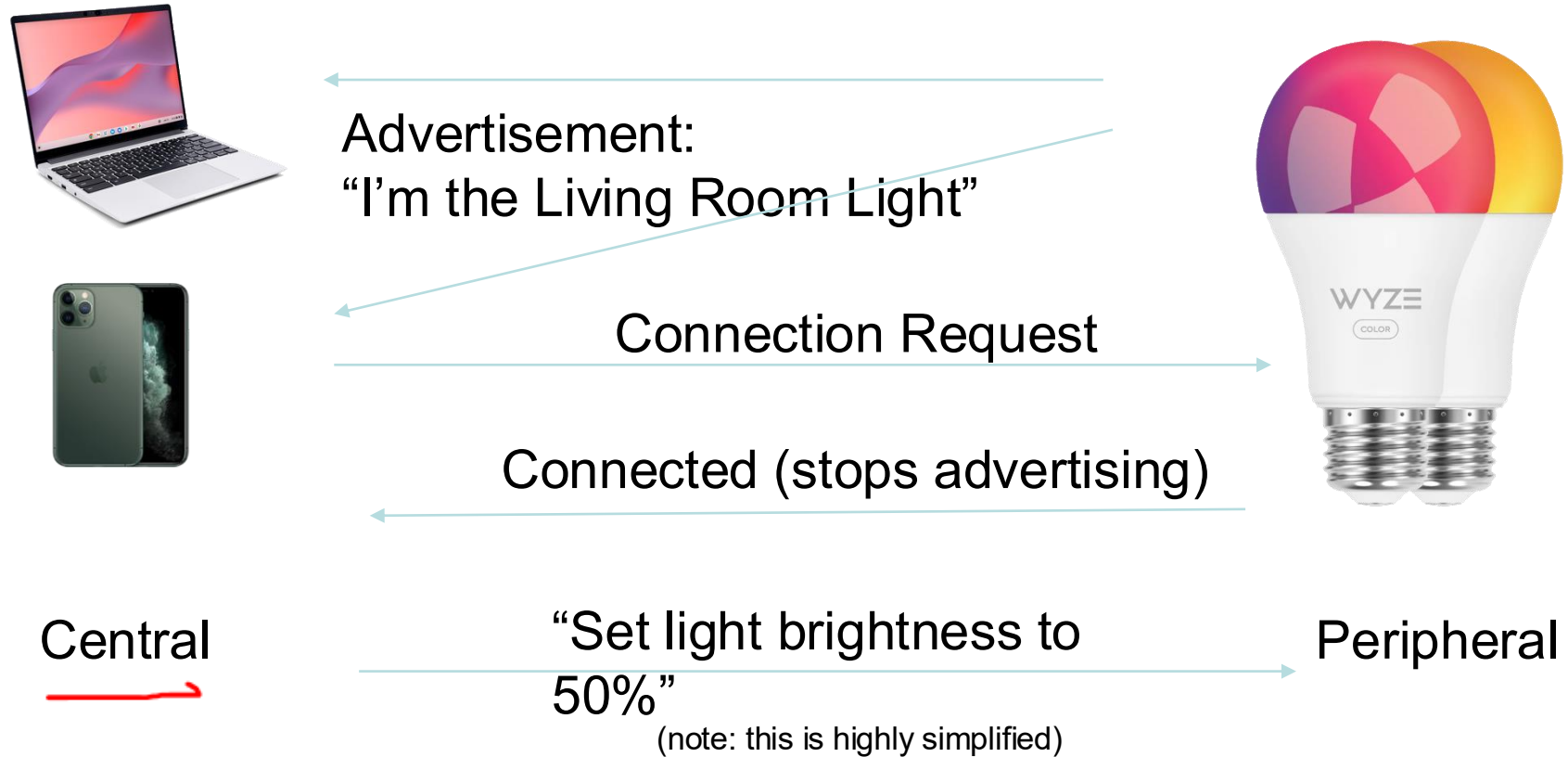
- Interaction
- Bi-directional communication between Peripheral and Central
- Maintained for some duration
- TDMA access control

# Bluetooth Low Energy Roles: Scanner, Advertiser

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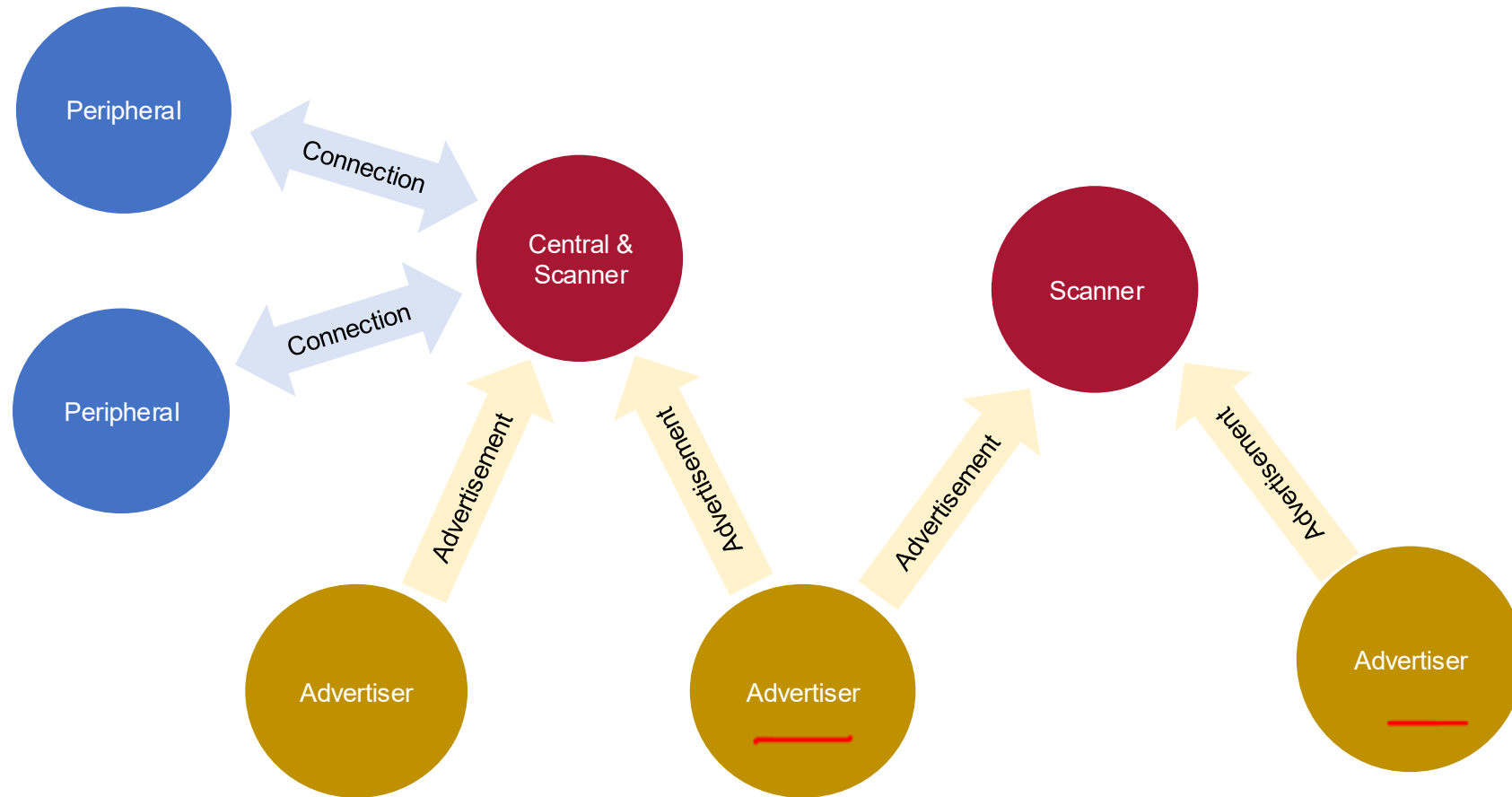


# BLE Central and Peripheral roles are for connections



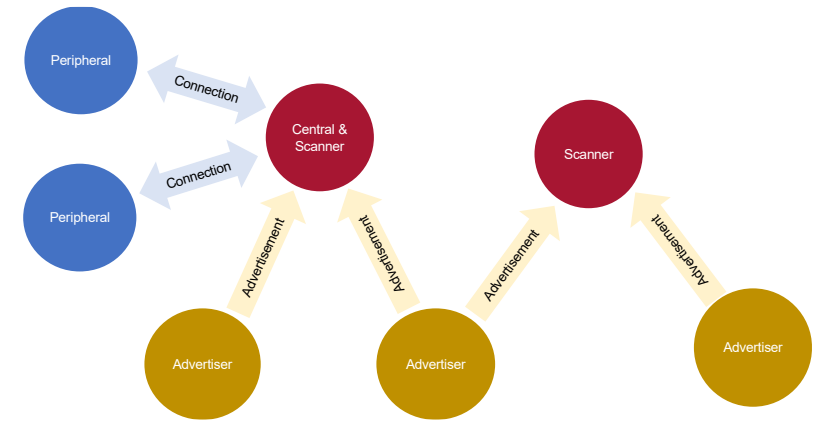


# BLE network topology



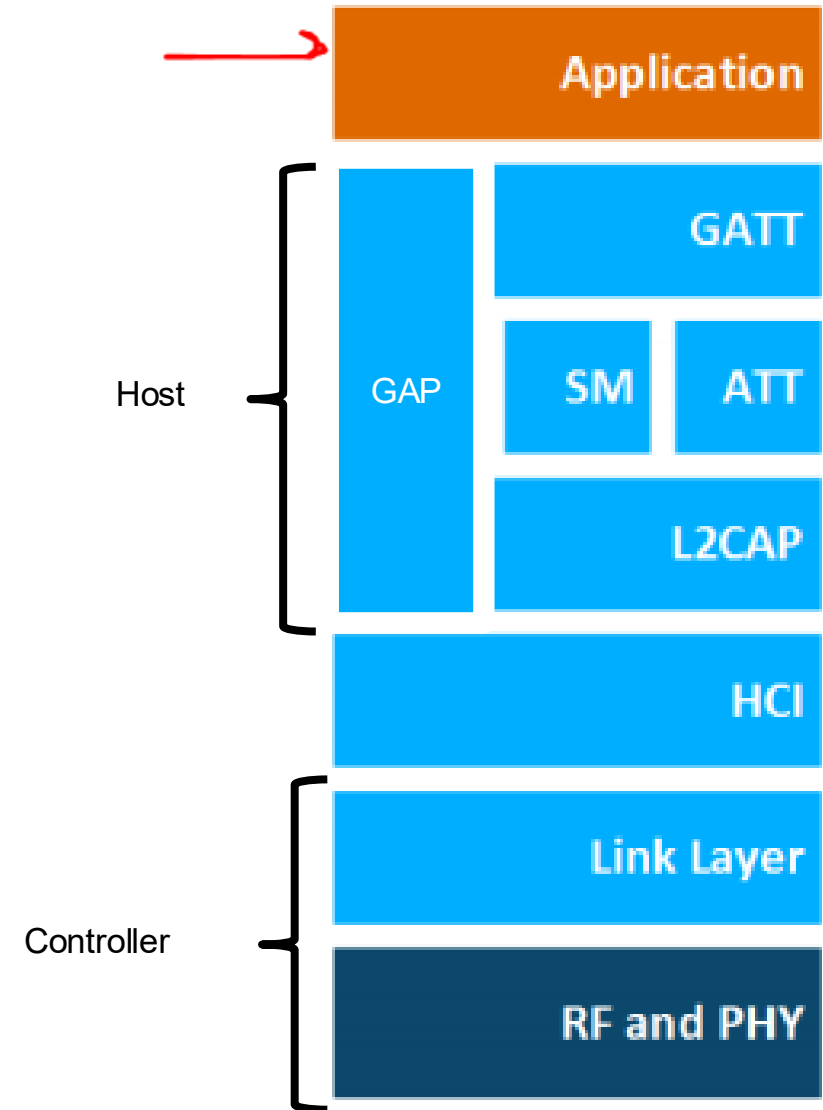
# Multiple roles at the same time

- Topology picture is a simplification of roles
- A single device can have multiple roles simultaneously
  - Scanning and Advertising simultaneously
  - Peripheral and Scanner and Advertiser simultaneously
  - Peripheral and Scanner and Central and Advertiser simultaneously
    - Getting a bit out of hand though



# BLE Layers

- Host – Configuration for advts, connections
  - GAP – Generic Access Profile
    - Configure advertising
  - GATT – Generic ATtribute profile
    - Configure connections
- HCI - Host Controller Interface
- Controller - Communication
  - Link Layer – send packets
  - RF and PHY – send bits



# Outline

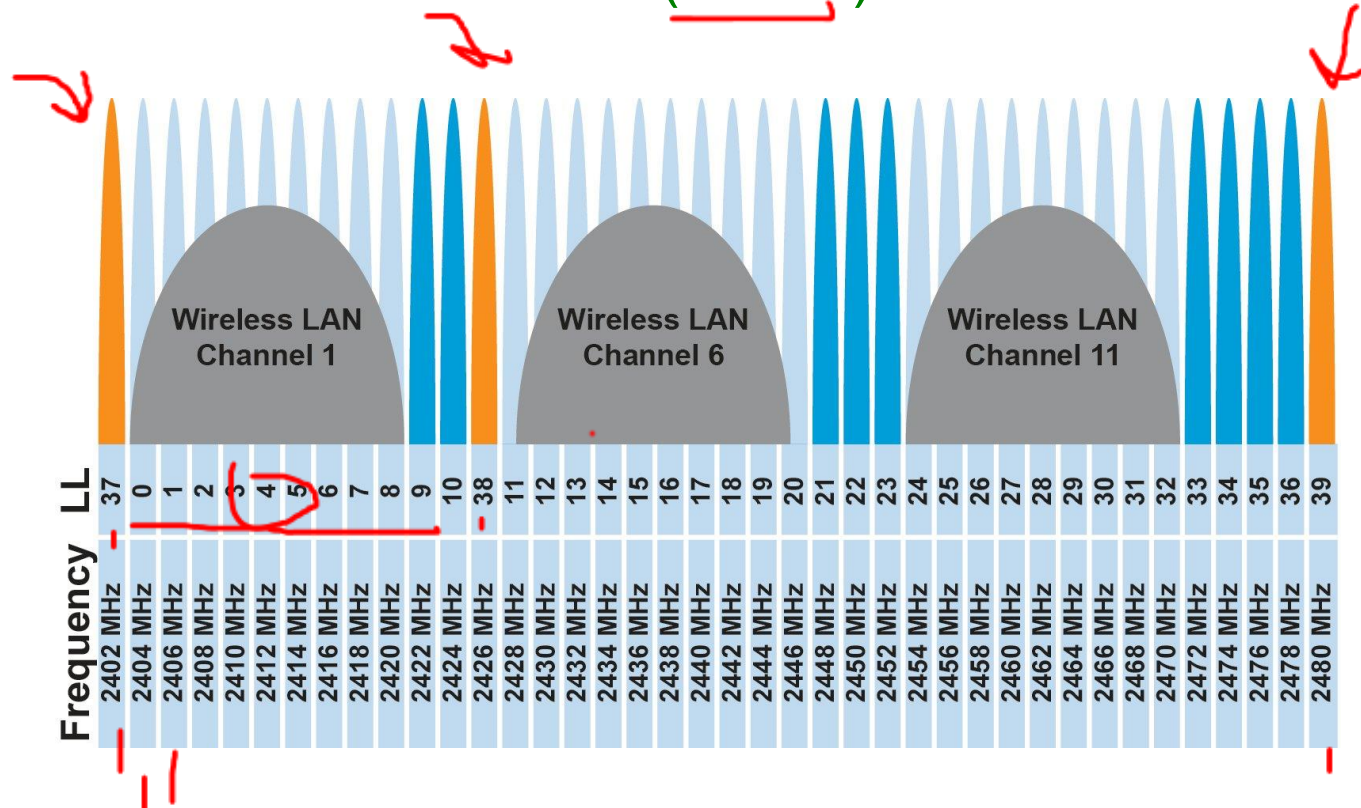
- BLE Background
- **BLE Layers**
  - Physical Layer
  - Link Layer
- BLE roles
  - Advertising
  - Scanning

# BLE frequency usage and channels

- 2.4 GHz carrier, forty 2-MHz channels, 1 Mbps data rate

→ 37, 38, 39 for advertising

- 0-36 for connection (FHSS)



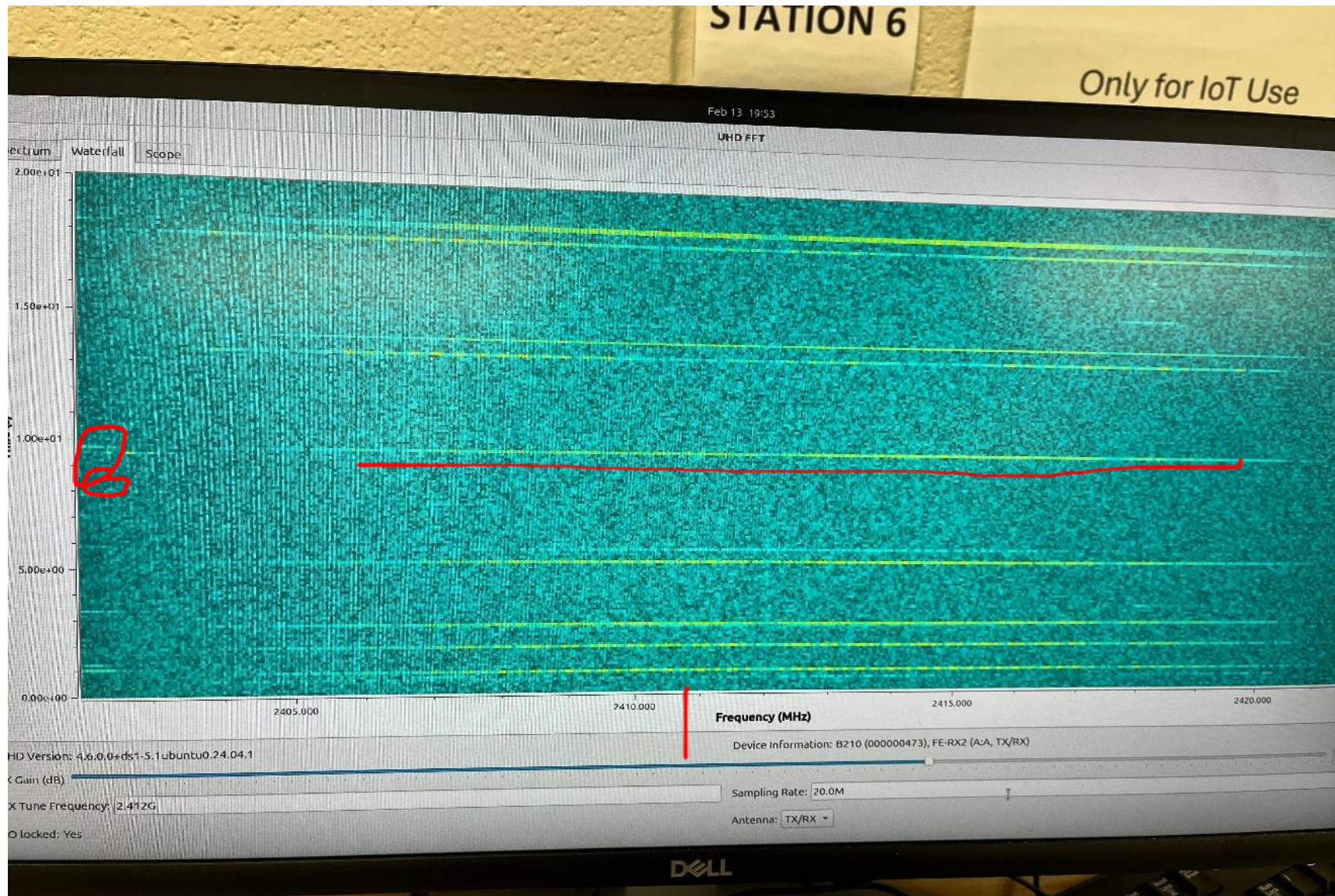
Why doesn't BLE avoid WiFi altogether?

Can't on 2.4 GHz

40

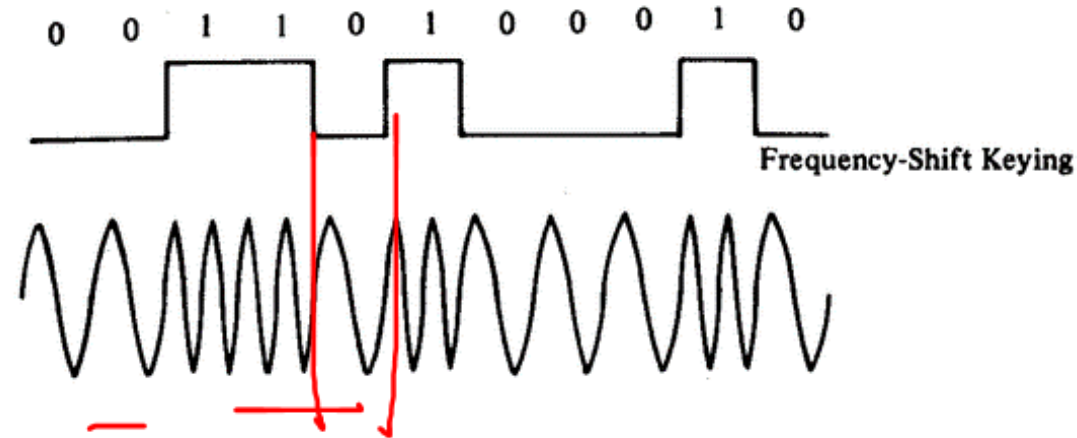


# Lab 3: Spectrum analysis for BLE with USRP setup

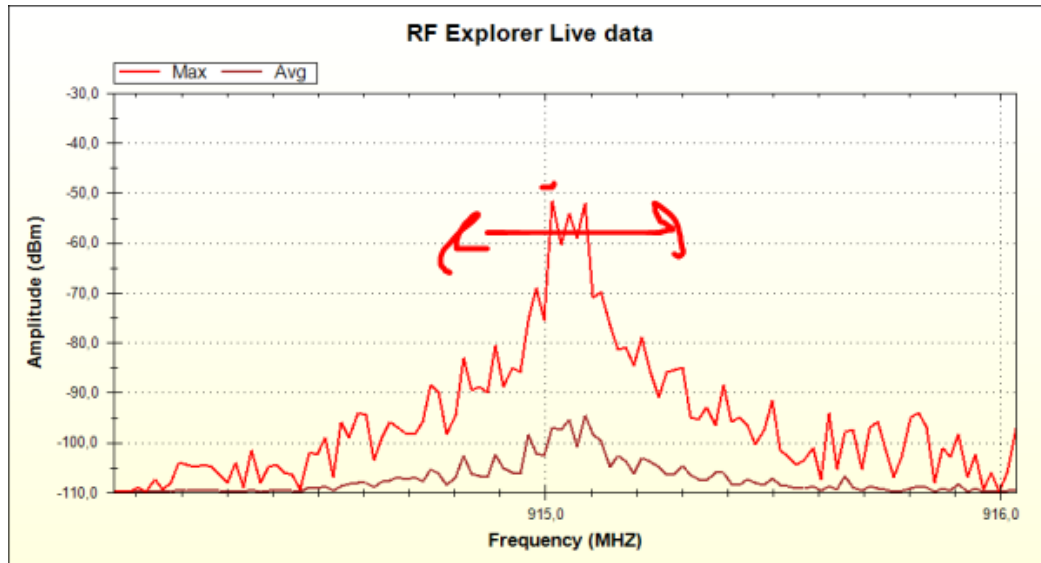


# BLE modulation

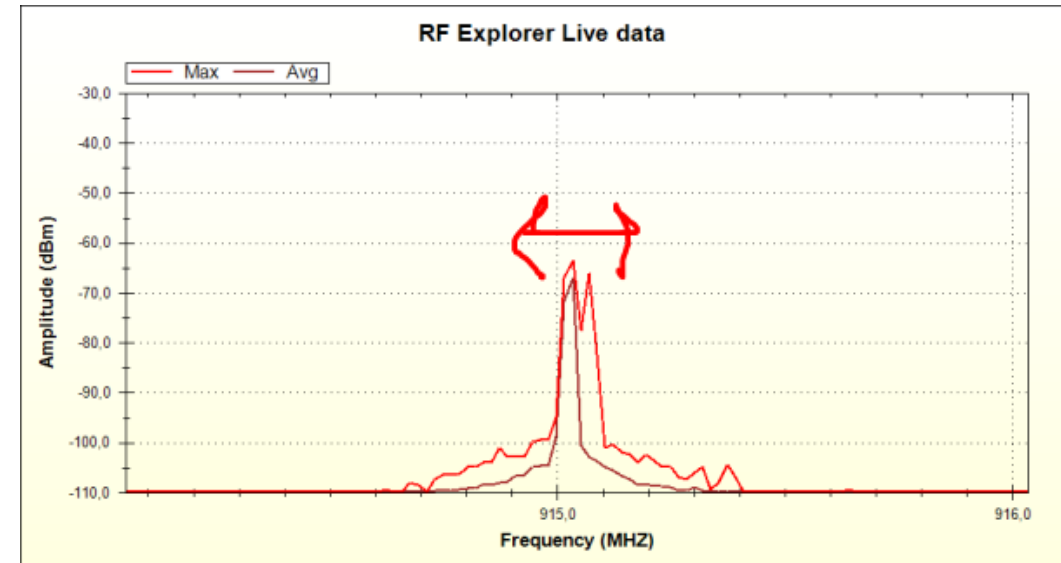
- Gaussian Frequency-Shift Keying (GFSK)
  - Improvement on base Frequency-shift Keying
  - Smoother transitions between bits -> reduces nearby interference



# Gaussian FSK lessens spectral leakage, reduces bandwidth



FSK



GFSK



# BLE signal strength

The requirements for a Bluetooth low energy radio are as follows:

| Feature                | Value              |
|------------------------|--------------------|
| Minimum TX power       | 0.01 mW (-20 dBm)  |
| Maximum TX power       | 100 mW (20 dBm)    |
| Minimum RX sensitivity | -70 dBm (BER 0.1%) |

The typical range for Bluetooth low energy radios is as follows:

| TX power | RX sensitivity | Antenna gain | Range      |
|----------|----------------|--------------|------------|
| 0 dBm    | -92 dBm        | -5 dB        | 160 meters |
| 10 dBm   | -92 dBm        | -5 dB        | 295 meters |

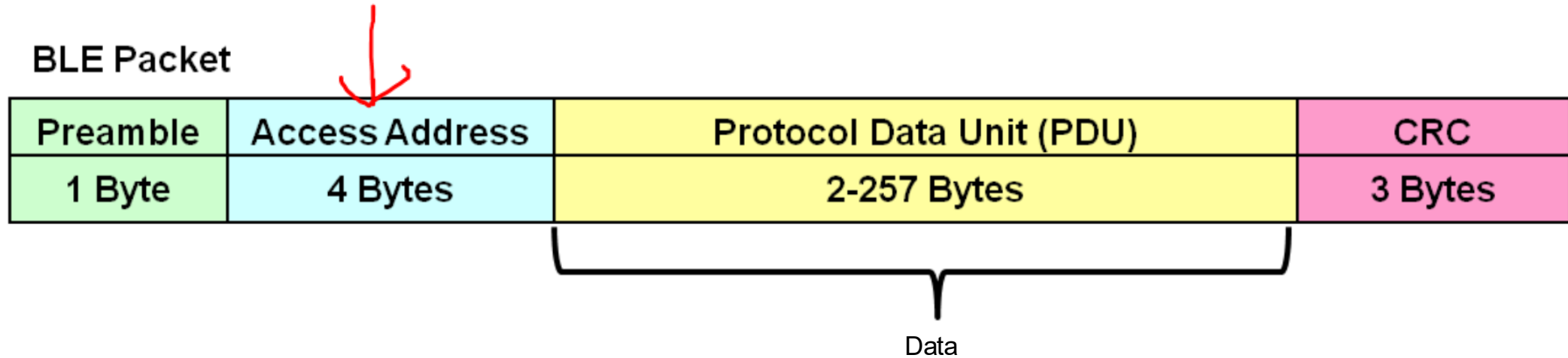
The range to a smart phone is typically 0-50 meters due to limited RF performance of the phones.

- nRF52840 (the SoC) capabilities
  - Transmit: up to 8 dBm
  - Receive sensitivity: -95 dBm

# Outline

- BLE Background
- **BLE Layers**
  - Physical Layer
  - **Link Layer**
- **BLE roles**
  - Advertising
  - Scanning

# Link Layer Packet Structure



- Same packet structure for both advertisements and connections
  - Fields are filled in little endian
- Access address unique for each connection (randomly chosen)
- All Advertisements use the same AA: 0x8E89BED6

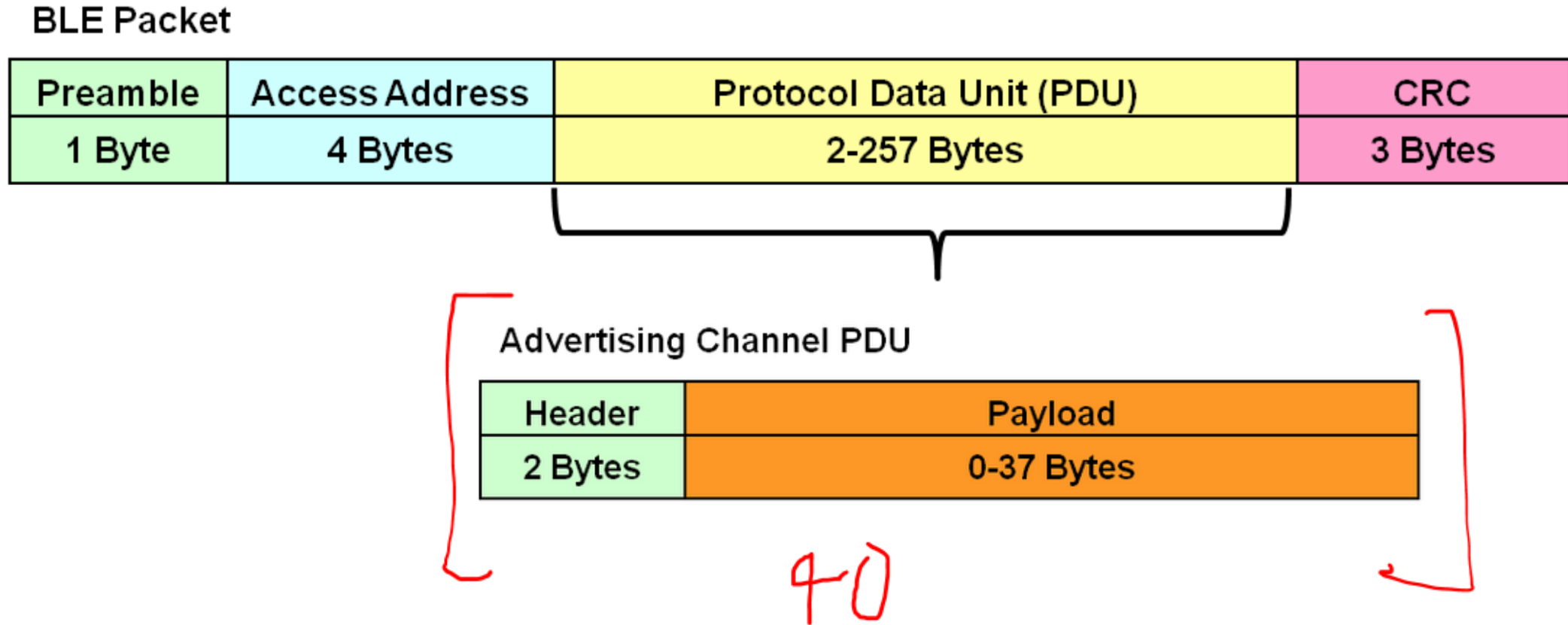
# Outline

- BLE Background
- BLE Layers
  - Physical Layer
  - Link Layer
- **BLE roles**
  - Advertising 
  - Scanning 

# Advertising: BLE's discovery mechanism

- Make nearby devices aware of advertiser's existence
- Communicate some information from or about advertiser
- Traditional purpose is to enable connections, but this is also useful for general communication
- Advertisements
  - Periodic broadcast messages with data
- Scan Requests/Responses
  - Scanner sends responses after getting a request
    - Only occurs when scanner is listening
  - Almost literally “bonus advertisement data”

# Advertising packet layering



# BLE advertising header

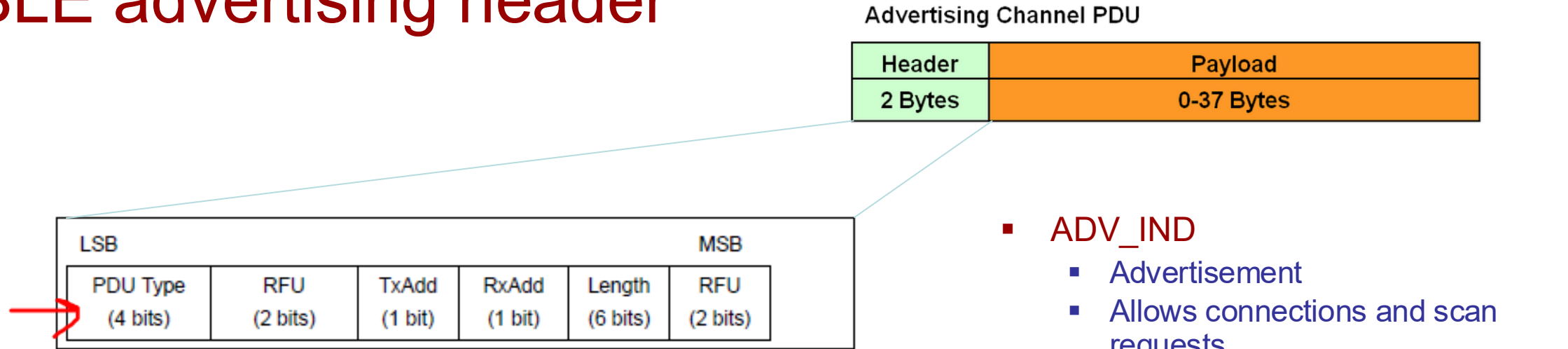


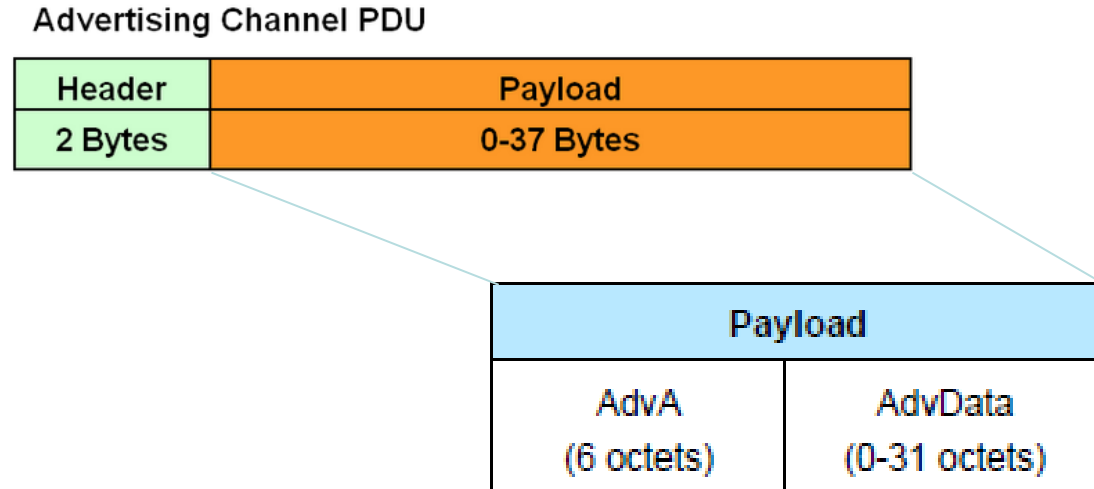
Figure 2.3: Advertising channel PDU Header

| PDU Type<br>$b_3b_2b_1b_0$ | Packet Name     |
|----------------------------|-----------------|
| 0000                       | ADV_IND         |
| 0001                       | ADV_DIRECT_IND  |
| 0010                       | ADV_NONCONN_IND |
| 0011                       | SCAN_REQ        |
| 0100                       | SCAN_RSP        |
| 0101                       | CONNECT_REQ     |
| 0110                       | ADV_SCAN_IND    |
| 0111-1111                  | Reserved        |

Table 2.1: Advertising channel PDU Header's PDU Type field encoding

- **ADV\_IND**
  - Advertisement
  - Allows connections and scan requests
- **ADV\_NONCONN\_IND**
  - Advertisement
  - No connections or scan requests
- **ADV\_SCAN\_IND**
  - Advertisement
  - No connections but allows scan requests
- **SCAN\_REQ**
  - Scan request
- **SCAN\_RSP**
  - Scan response

# Advertisement payloads



- AdvA – address of the advertiser
  - TxAdd bit from header specifies if this is a “public” or “random” address
- Remaining up to 31 bytes are available for use

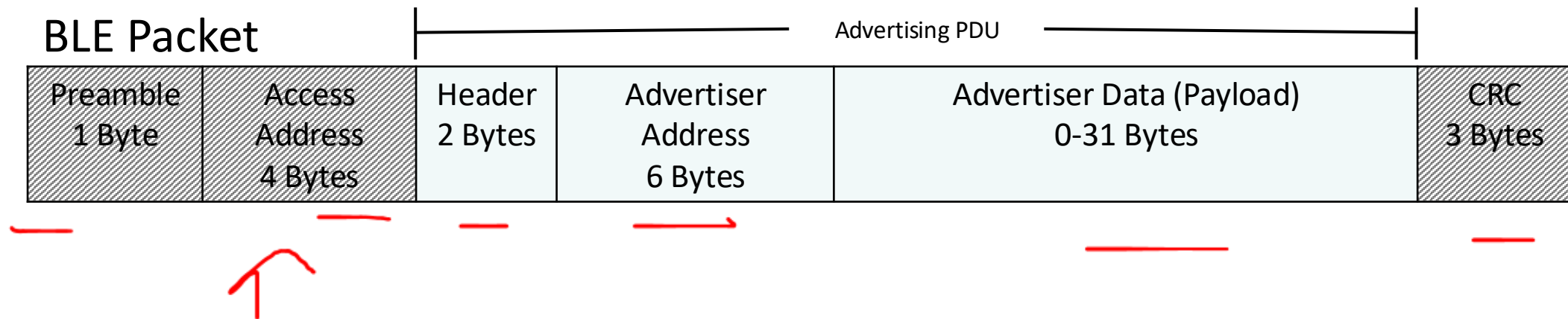


# Advertiser addresses

- Public and private address forms
- Public
  - 48 bits: 24-bits of company ID, 24-bits of company assigned number
  - Literally the same MAC address scheme as Ethernet and WiFi
- Private
  - Top two MSbs specify type
    - 46 bits of random
    - 46 bits of hash of an identity key
- Why have the two types?
  - Privacy

# Advertisement Packet

- Putting it all together, up to 47 bytes total:



# Scan Requests and Responses

- Scan request
  - Just the two addresses: the scanner's and the advertiser's

| Payload             |                    |
|---------------------|--------------------|
| ScanA<br>(6 octets) | AdvA<br>(6 octets) |

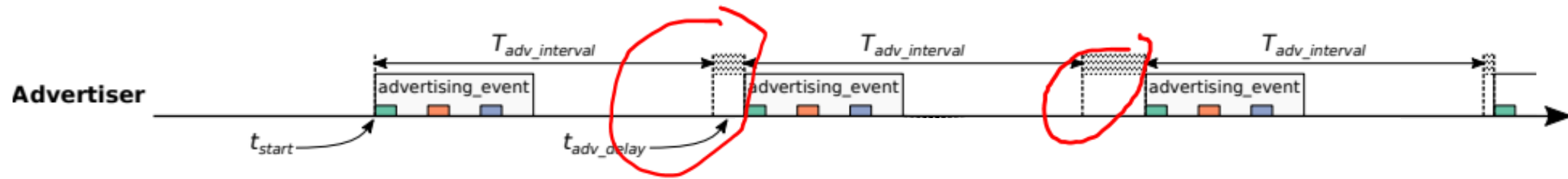
Figure 2.8: SCAN\_REQ PDU Payload

- Scan response
  - Identical to an advertisement
  - But only occurs after a request

| Payload            |                              |
|--------------------|------------------------------|
| AdvA<br>(6 octets) | ScanRspData<br>(0-31 octets) |

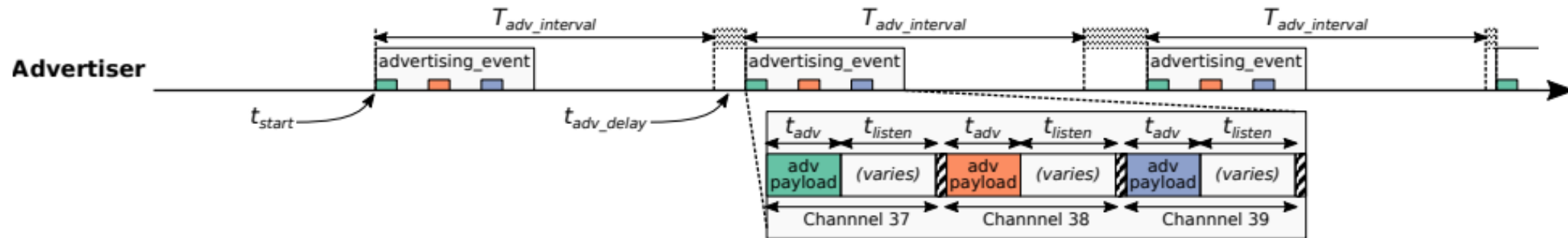
Figure 2.9: SCAN\_RSP PDU payload

# Advertising timing



- Advertising Events occur periodically [20ms – 10.24 s] (or longer)
  - User picks the rate as a tradeoff of energy and discovery latency
- Plus a random delay after each instance [0-10ms]
  - **Why?**
    - Avoid repeat collisions.

# Advertising event



- Three transmissions, one on each advertising channel
  - Always in the same order
- Transmission, followed by listening window on that same channel
  - Requests will be sent  $\geq 150$  us (Inter-Frame Spacing, IFS) after Tx
  - Followed by a retune to the next channel frequency
- This short listen window is the magic “low energy” part

# Preserving energy in communication

- Most energy is spent listening
  - This is due primarily to how long listening durations are compared to transmissions
- Question: advertising vs. listening in BLE:
  - How much more energy is used listening for 1 second versus transmitting 30 bytes three times?
    - Radio uses 60 mW when active (TX or RX)
    - 1 Mbps data rate

# Preserving energy in communication

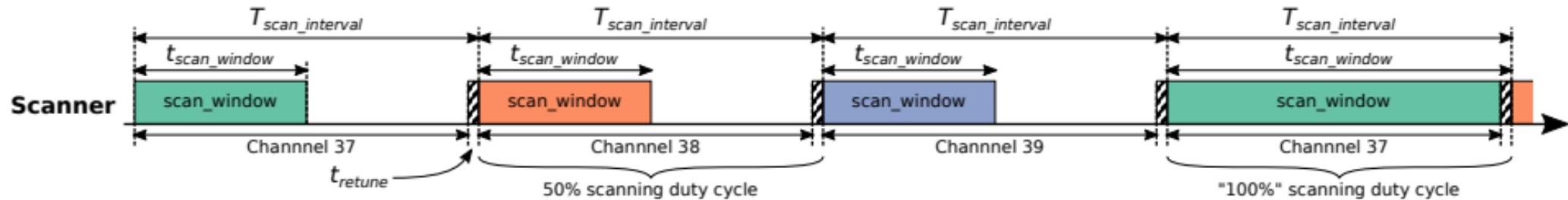
- 30 bytes = 240 bits
- $240 \text{ bits} / 1 \text{ Mbps} = 0.240 \text{ ms}$
- $0.24 * 3 = 0.72 \text{ ms}$  (three channels)
- $1 \text{ s} / 0.720 \text{ ms} = 1389$
- 1389x!!

# Outline

- BLE Background
- BLE Layers
  - Physical Layer
  - Link Layer
- BLE roles
  - Advertising
  - **Scanning**



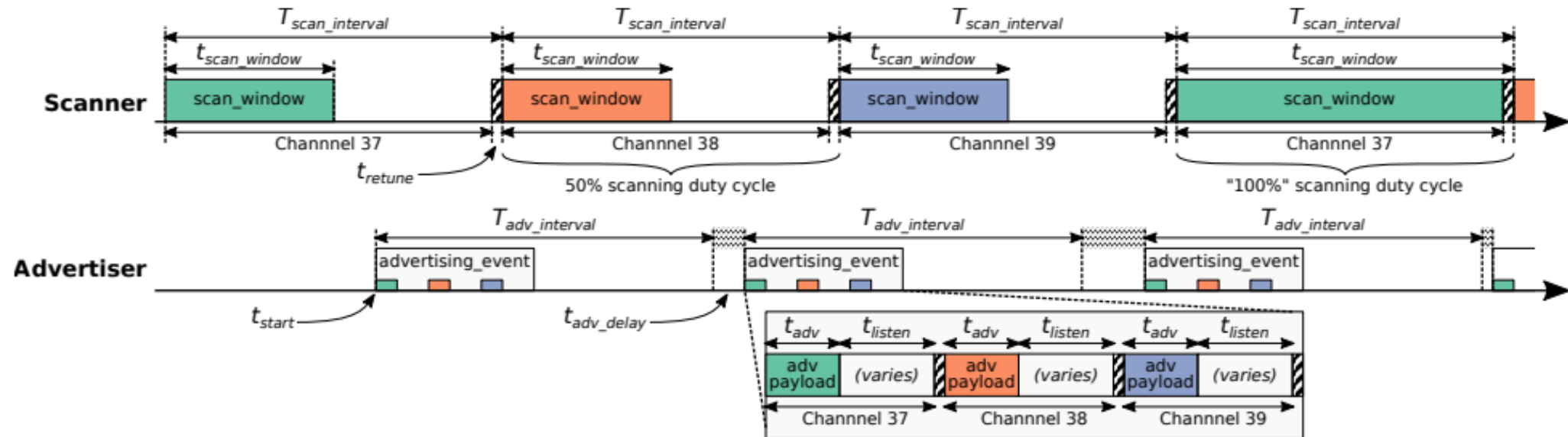
# Scanning Pattern



- Iterate through channels, listening for advertisements
  - $T_{\text{scan\_interval}}$  controls rate at which channels are changed
  - $T_{\text{scan\_window}}$  controls duty cycle of listening
- Why listen at a low duty cycle?
  - Save energy

# Putting it all together

- Advertisements are received when the channel of the scan window and the channel of the advertisement overlap in time (and space)

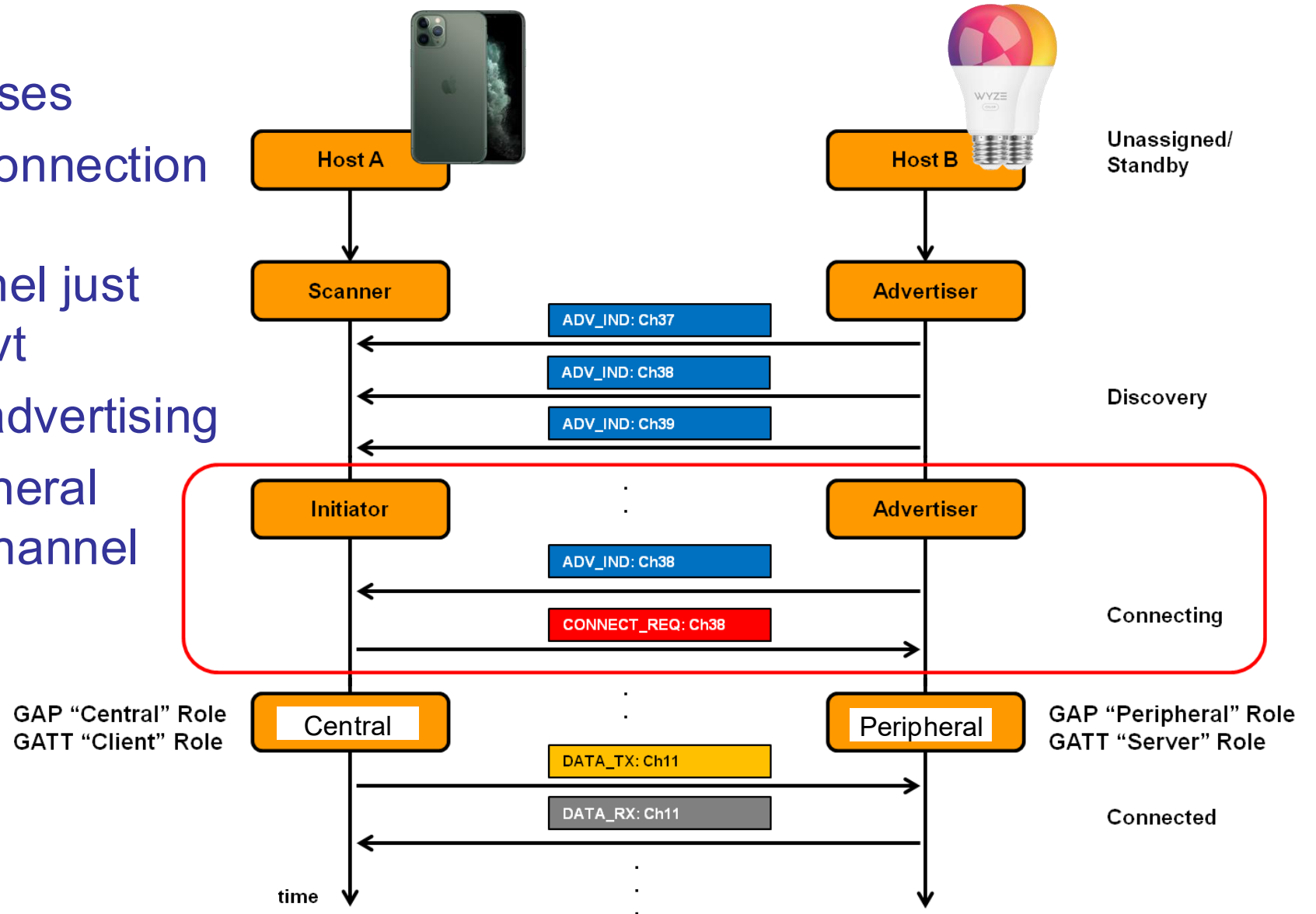


# A warning about scanning expectations

- Scanners will NOT receive 100% of packets sent
  - Even ignoring range issues
- Packets are lost due to (in roughly descending order):
  - Duty cycle
  - Sharing 2.4 GHz antenna with WiFi
  - Retune period after each scanning interval
  - Dropped packets in the receive software
  - Packet collisions

# Connection Timeline

- Peripheral advertises
- Central sends a connection request packet on advertising channel just after receiving advt
- Peripheral stops advertising
- Central and peripheral switch to a data channel



# Lab 3: BLE packet analysis tool

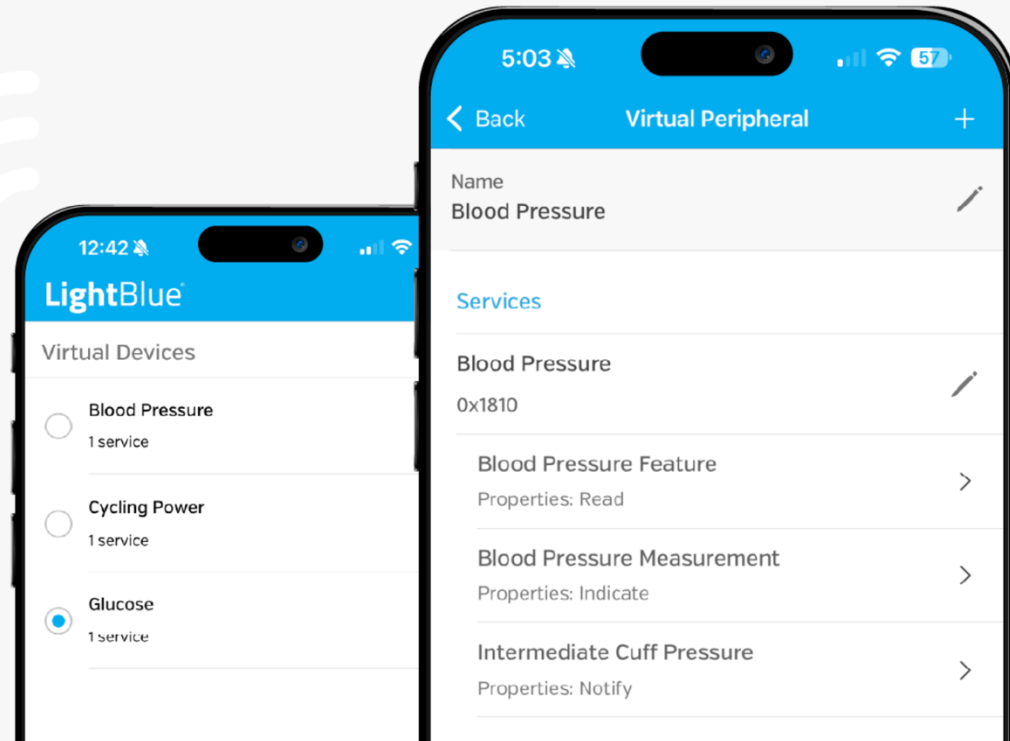
LIGHTBLUE®

## Your Ultimate BLE Debugging & Development Toolkit

For Mobile & Embedded Software Developers



Available on iOS, macOS, and Android.



Next topic: IEEE 802.15.4 and Zigbee